		a
19	**A**	Time proton spends between the parallel plates, $t = \dfrac{y}{v}$ Vertical acceleration, $a_{\mathrm{y}} = \dfrac{F_{\mathrm{E}}}{m} = \dfrac{eE}{m} = \dfrac{eV}{md}$ Using $s_{\mathrm{y}} = u_{\mathrm{y}}t + \dfrac{1}{2}a_{\mathrm{y}}t^2$, $x = \dfrac{1}{2}a_{\mathrm{y}}t^2$ (no initial vertical velocity) $= \dfrac{1}{2}\left(\dfrac{eV}{md}\right)\left(\dfrac{y}{v}\right)^2 = \dfrac{eVy^2}{2mdv^2}$ Alternatively, can also use unit analysis: $[eV] = \mathrm{kg\ m^2\ s^{-2}}$ Option A: $[\dfrac{eVy^2}{2mdv^2}] = \dfrac{\mathrm{kg\ m^2\ s^{-2}\ m^2}}{\mathrm{kg\ m\ m^2\ s^{-2}}} = \mathrm{m}$ Option B: $[\dfrac{2mdv^2}{eVy^2}] = \dfrac{\mathrm{kg\ m\ m^2\ s^{-2}}}{\mathrm{kg\ m^2\ s^{-2}\ m^2}} = \mathrm{m^{-1}}$ Option C: $[\dfrac{eVv^2}{2mdy^2}] = \dfrac{\mathrm{kg\ m^2\ s^{-2}\ m^2\ s^{-2}}}{\mathrm{kg\ m\ m^2}} = \mathrm{m\ s^{-4}}$ Option D: $[\dfrac{mdy^2}{2eVv^2}] = \dfrac{\mathrm{kg\ m\ m^2}}{\mathrm{kg\ m^2\ s^{-2}\ m^2\ s^{-2}}} = \mathrm{m^{-1}\ s^4}$
20	A	The wire carries a constant electric current I